Chapter 7. Complete Metric Spaces and Function Spaces

Completeness and completion, compactness in metric spaces, compact convergence, Ascoli's theorem

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7.1 Complete metric spaces

A metric space is **complete** if every Cauchy sequence converges. A Cauchy sequence with a convergent subsequence converges; closed subsets of complete spaces are complete. $\mathbb{R}^k$ is complete (Cauchy sequences are bounded, and Bolzano–Weierstrass applies).

Theorem 7.1

(a) $\mathbb{R}^\omega$ is complete in the metric D of Theorem 2.6. (b) If Y is complete, then Y^J is complete in the uniform metric $\bar{\rho}$. (c) For any space X, the bounded continuous functions $\mathcal{C}(X, Y)$ and the continuous functions with $\bar{\rho}$ form closed subspaces of Y^X, hence complete metric spaces.

Proof. (b) A $\bar{\rho}$-Cauchy sequence is uniformly Cauchy, hence converges coordinatewise, uniformly. (c) Uniform limits of continuous functions are continuous. ■

A metric space can be complete in one metric and not in an equivalent one ($\mathbb{R}$ versus $(-1, 1)$), so completeness is not a topological property.

Theorem 7.2 (Completion)

Every metric space (X, d) isometrically imbeds as a dense subset of a complete metric space $\tilde{X}$, unique up to isometry.

Proof. Fix x_0 and map $x \mapsto \phi_x$, $\phi_x(y) = d(y, x) - d(y, x_0)$, into the complete space of bounded functions with the sup metric; $\|\phi_x - \phi_{x'}\|_\infty = d(x, x')$. Take the closure of the image. ■

Example 7.1 (Peano space-filling curve)

Define piecewise linear paths $f_n : [0, 1] \to [0, 1]^2$ by repeatedly replacing each segment by a zigzag through the four quarter-squares. Then $|f_{n+1} - f_n| \le \sqrt{2}/2^n$ uniformly, so $f_n \to f$ uniformly by completeness of $\mathcal{C}([0, 1], [0, 1]^2)$, and f is continuous; every point of the square is within $\sqrt{2}/2^n$ of $f_n([0, 1])$, so f is onto (its image is compact and dense).

7.2 Compactness in metric spaces

A metric space is **totally bounded** if for every $\epsilon > 0$ it is covered by finitely many ϵ-balls.

Theorem 7.3

A metric space is compact iff it is complete and totally bounded.

Proof. Compact implies these (cover by all ϵ-balls; a Cauchy sequence has a convergent subsequence). Conversely, given a sequence, total boundedness lets us choose, by a diagonal argument, nested subsequences lying in balls of radius $1/k$; the diagonal is Cauchy, hence convergent, so the space is sequentially compact, hence compact (Theorem 3.11). ■

Theorem 7.4 (Ascoli, classical form)

Let X be compact and $\mathcal{F} \subset \mathcal{C}(X, \mathbb{R}^n)$ with the sup metric. Then $\overline{\mathcal{F}}$ is compact iff $\mathcal{F}$ is equicontinuous and pointwise bounded.

Proof. If $\mathcal{F}$ is equicontinuous and pointwise bounded, it is totally bounded: given ϵ, cover the compact X by finitely many δ-neighbourhoods of points $a_1, \dots, a_k$ (from equicontinuity); the values at the a_i lie in a bounded set covered by finitely many small balls, and functions with close values

at all a_i are uniformly 3ϵ-close. The closure of an equicontinuous family is equicontinuous, and $\mathcal{C}$ is complete, so $\overline{\mathcal{F}}$ is complete and totally bounded. The converse follows from total boundedness of compact sets. ■

7.3 Pointwise and compact convergence

On Y^X the **topology of pointwise convergence** is the product topology. The **topology of compact convergence** has basis sets $B_C(f,\epsilon) = \{g : \sup_C d(f,g) < \epsilon\}$ for compact $C \subset X$; $f_n \to f$ in it iff $f_n \to f$ uniformly on compact sets. The **compact-open topology** on $\mathcal{C}(X,Y)$ has subbasis $S(C,U) = \{f : f(C) \subset U\}$ (C compact, U open); for metric Y it coincides on $\mathcal{C}(X,Y)$ with the topology of compact convergence, so it depends only on the topology of Y.

Theorem 7.5

If X is locally compact Hausdorff, the evaluation map $e : X \times \mathcal{C}(X,Y) \to Y$, $e(x,f) = f(x)$, is continuous (compact-open topology). Moreover a map $F : X \times Z \to Y$ is continuous iff its adjoint $Z \to \mathcal{C}(X,Y)$, $z \mapsto F(\cdot, z)$, is continuous.

The second statement means that homotopies $X \times I \to Y$ are the same as paths in $\mathcal{C}(X,Y)$ — a viewpoint used in Part II.

7.4 Ascoli's theorem

Theorem 7.6 (Ascoli)

Let X be locally compact Hausdorff and (Y,d) a metric space; give $\mathcal{C}(X,Y)$ the topology of compact convergence. A subset $\mathcal{F}$ has compact closure if $\mathcal{F}$ is equicontinuous under d and $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ has compact closure for each $a \in X$. The converse holds if $Y = \mathbb{R}^n$.

Idea. Imbed $\mathcal{F}$ in the product $\prod_a \overline{\mathcal{F}_a}$, compact by Tychonoff. On equicontinuous families the product topology (pointwise convergence) and the compact convergence topology agree, and the closure of $\mathcal{F}$ in the product consists of continuous functions (equicontinuity passes to pointwise limits). ■

Example 7.2

A sequence of differentiable $f_n : [0,1] \to \mathbb{R}$ with $|f_n(0)| \le 1$ and $|f_n'| \le 1$ has a uniformly convergent subsequence. Families of holomorphic functions bounded on compact sets are equicontinuous (Cauchy estimates), giving Montel's theorem in complex analysis.